

ASSIGNMENT 2

Research for Engineering Decisions

Assessment type	Team research assignment
Team size	3 students - same team used for the integrated project
Duration	Weeks 3-4; research is used progressively as relevant decisions are made
Total marks	50
Submission	One concise professional engineering research brief per team that is properly formatted with a coverage. NB: Include a list of team members and mention those who were not active.
Project relationship	Research feeds relevant CivicConnect Milestone 2 decisions; Assignment 2 does not replace project documentation
Referencing	Havard academic referencing style
Academic year	2026

Design • Integration • Collaborative Engineering • Continuous Integration

Group Assignment | Same CivicConnect Teams | 50 Marks | Weeks 3–4 Research Feeding Project Decisions

Central Purpose

Assignment 2 is the research input to the team's developing CivicConnect solution. Teams investigate engineering concepts, compare credible alternatives and make evidence-based recommendations BEFORE or while the corresponding project decisions are being baselined. Milestone/project documentation then records the team's final project-specific decision, rationale, traceability and application evidence.

1. How Assignment 2 (A2) and Milestone 2 (M2) Work Together

Assignment 2 and Milestone 2 are related but they do not assess the same evidence. Assignment 2 assesses the quality of research, comparison and recommendation. Milestone 2 assesses how the team converts relevant research into controlled engineering decisions and begins applying those decisions in CivicConnect.

Research Question → Credible Evidence → Alternatives → Comparison → Recommendation → CivicConnect Decision → PED/ADR/RTM → Application Evidence

Do the research before an unsupported commitment

Where Assignment 2 research is relevant to an M2 decision, use the research to inform the decision before baselining it. Do not first make an arbitrary project choice and then search for sources that agree with it. Research informs judgement; it does not replace judgement.

2. Progressive Research — Do Not Wait for the Whole Assignment

This assignment spans Weeks 3–4. Teams should work on each research section as the corresponding concept becomes relevant in class and in the developing project. Findings that are needed for an immediate project decision should be used when that decision is made; teams do not need to wait for the final Assignment 2 submission date.

- Week 3 research primarily supports internal design, persistence and integration decisions.
- Week 4 research primarily supports collaborative engineering, configuration management, Continuous Integration and automated verification decisions.
- If a project decision must be made before a later research section is completed, document the current evidence, assumptions and risk; refine the decision through controlled change when stronger evidence becomes available.
- Assignment findings that affect the project must be reflected in the existing Project Engineering Document (PED), Requirements Traceability Matrix (RTM), Architecture Decision Records (ADRs), and relevant application documentation. Do not create a second parallel project record inside the assignment.

3. Research Standard

This is not a definitions assignment. For each task, research must lead to a recommendation that could genuinely influence CivicConnect. Use credible academic/professional sources for software-engineering concepts and authoritative/official sources for technology-specific claims.

- Explain the engineering problem before discussing a technique or pattern.
- Compare alternatives, including a simpler approach where appropriate.
- Evaluate benefits AND costs/limitations/consequences.

- Relate the evidence to CivicConnect requirements, ASRs, architecture, data, team capability, risks and constraints.
- Distinguish what the sources say from what your team recommends.
- Use consistent in-text citations and a complete reference list.
- Verify AI-assisted claims, current versions, compatibility, licensing and security information against authoritative sources.

4. TASK 1 — Researching Design Quality and Design Patterns

Your team is beginning implementation. Before design structures become expensive to change, research how internal design choices can control coupling, improve cohesion and support maintainability, extensibility and testability.

4.1 Establish the design problem

- Explain coupling and cohesion and why they matter once multiple classes/components must change together.
- Research the SOLID principles most relevant to the kinds of design problems emerging or anticipated in CivicConnect. You are not required to reproduce all five if they do not contribute to the analysis.
- Identify at least TWO genuine CivicConnect design problems that should be addressed before substantial implementation. Describe the problem without naming a pattern first.

4.2 Research alternatives for each problem

For EACH of the two design problems, research at least TWO plausible design patterns/approaches. Suitable patterns may include Factory, Adapter, Observer or other recognised approaches relevant to the actual problem; the assignment does not prescribe which patterns the team must choose.

- Explain the intent and structure/responsibility distribution of each candidate.
- Evaluate effects on coupling, cohesion, maintainability, extensibility and testability.
- Identify complexity, misuse and over-engineering risks.
- Compare when each candidate is appropriate and when a simpler solution would be preferable.

4.3 Recommendation feeding M2

For each design problem, recommend the pattern/approach that your research indicates is most appropriate for CivicConnect. Your recommendation becomes evidence that the team may use when making the formal M2 design decision.

A2 recommends; M2 commits

Assignment 2 does NOT earn marks merely because M2 uses the recommendation. A2 earns marks for the research and comparison. In M2 the team must still make and document its final project-specific choice of at least two justified patterns/approaches, record the rationale/ADR and show where the decision will be applied. The final M2 choice may differ from the A2 recommendation if the team has defensible project-specific evidence.

TASK 1 — 15 Marks

Criterion	Marks	What earns the marks
Design-quality foundation and problem identification	4	Accurate coupling/cohesion/relevant SOLID reasoning; two genuine problems identified before naming solutions.
Research and comparison of candidate patterns/approaches	7	At least two credible alternatives per problem; structure, benefits, limitations, misuse and quality consequences evaluated.
Evidence-based CivicConnect recommendations	4	Two justified recommendations explicitly showing how the research should inform M2 decisions.

5. TASK 2 — Researching Persistence and Data-Integrity Decisions

M2 establishes the data/persistence baseline, but implementation introduces deeper correctness questions. Research the mechanisms that should guide how CivicConnect protects important data as development begins.

5.1 Choose one meaningful business operation

Select one CivicConnect operation where incorrect or partially completed data would have a real consequence, for example request creation, assignment, status transition or history/audit recording.

5.2 Research the engineering choices

- Research transaction/atomicity requirements relevant to the operation.
- Research where validation and business rules can be enforced: UI/client, application/service and database — and the consequences of relying on only one layer.
- Investigate relevant consistency, concurrency and integrity mechanisms.
- Where performance techniques such as caching are plausible, investigate the correctness/staleness trade-off; do not introduce them without evidence.
- Compare at least two plausible implementation approaches or responsibility allocations.

5.3 Recommendation feeding the project

Recommend how the selected operation should preserve correctness and where the relevant responsibilities should sit. Identify which M2 data/persistence decision, risk, ADR, data model or implementation artefact the recommendation should inform.

TASK 2 — 10 Marks

Criterion	Marks	What earns the marks
Research into correctness mechanisms	4	Transactions, validation, integrity/consistency/concurrency accurately investigated for the chosen operation.
Alternative approaches and trade-offs	3	Meaningful alternatives compared rather than one preferred mechanism described.
CivicConnect recommendation and M2 link	3	Evidence-based recommendation identifies which project decision/artefact it should inform.

6. TASK 3 — Researching APIs and Integration Decisions

As modules/components begin collaborating, the team must decide how responsibilities communicate without creating uncontrolled coupling. Research an integration problem that CivicConnect is likely to face.

6.1 Identify an integration problem

- Choose one meaningful interaction between modules/components or a plausible external integration.
- Define the producer/caller, consumer/callee, information exchanged and failure consequences.
- Explain why the interaction requires an explicit engineering decision rather than an accidental dependency.

6.2 Compare integration approaches

- Research at least TWO plausible mechanisms, such as an in-process interface, HTTP/REST API, asynchronous event/message or another justified approach.
- Compare coupling, performance, failure behaviour, testability, deployment implications, security boundary and change/version compatibility.
- Where REST is a candidate, research resource/method/status/error/statelessness principles sufficiently to evaluate it; do not assume every internal interaction needs HTTP.
- Consider the cost of distribution. A network boundary must be justified.

6.3 Recommendation feeding the project

Recommend the most proportionate mechanism for the chosen CivicConnect interaction and identify the architecture/design/interface/ADR/application documentation that should reflect the eventual project decision.

TASK 3 — 10 Marks

Criterion	Marks	What earns the marks
Integration problem framing	2	Responsibilities, exchanged information and failure/change consequences are clear.
Research and comparison of mechanisms	5	At least two plausible mechanisms evaluated across relevant engineering consequences.
CivicConnect recommendation and project link	3	Proportionate recommendation and affected project evidence clearly identified.

7. TASK 4 — Researching Collaborative Engineering and CI Controls

A three-person team can begin coding quickly, but uncontrolled integration creates defects, merge problems and unverified changes. Research the controls that should govern how your team integrates work as development increases.

7.1 SCM and collaborative integration

- Research the difference between version control and broader Software Configuration Management.
- Investigate branching, Pull Request, peer-review and traceability practices suitable for a small engineering team.
- Use the Master Project Brief's protected-main and two-reviewer requirements as project constraints, then research how the workflow can satisfy them meaningfully rather than mechanically.
- Identify at least one likely collaboration/integration risk in CivicConnect and compare ways to control it.

7.2 Continuous Integration and quality gates

- Research the purpose of Continuous Integration and distinguish it from simply using Git or having an automated deployment pipeline.
- Investigate what should trigger CI and what a repeatable build requires for your likely/selected stack.
- Research appropriate automated checks: compilation/build, unit tests, lint/static analysis, dependency/security checks or other stack-appropriate verification.
- Compare which checks should block integration and which may initially provide warnings, with reasons.
- Research secrets/configuration handling and how CI results should become visible to reviewers.
- Explain why automation supports but does not replace human peer review.

7.3 Recommended team control

Propose a research-informed collaborative integration and CI approach for CivicConnect. This recommendation should feed the team's repository/process decisions as development progresses. A full mature CI/CD implementation is not required by the research assignment.

TASK 4 — 12 Marks

Criterion	Marks	What earns the marks
SCM/collaboration research	4	VCS vs SCM, review/traceability practices and project risk supported by credible evidence.
CI/quality-gate research and alternatives	5	Triggers, build, checks, blocking rules, secrets/results and human review critically evaluated.
CivicConnect workflow recommendation	3	Practical, proportional recommendation aligned to the Master Project Brief and team context.

8. TASK 5 — Research-to-Decision Map

Create one concise table that demonstrates how Assignment 2 findings are intended to feed the project. This is NOT a duplicate PED or RTM; it is the bridge from research evidence to engineering decision-making.

A2 finding	Evidence/alternative considered	Recommendation	Project decision it should inform	Expected PED/ADR/RTM/application evidence
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Design problem 1

Design problem 2

Persistence/data

Integration/API

SCM/CI

TASK 5 — 3 Marks

Marks reward clear traceability from research to recommendation to the correct project decision/evidence. Copying assignment paragraphs into the table does not earn additional credit.

9. Mark Summary

Assessment area	Marks
Task 1 — Design Quality & Design Patterns	15
Task 2 — Persistence & Data Integrity	10
Task 3 — APIs & Integration	10
Task 4 — Collaborative Engineering & CI	12
Task 5 — Research-to-Decision Map	3
TOTAL	50

10. Required Submission

- One coherent professionally formatted group Assignment 2 research report covering Tasks 1–5.
- Consistent in-text citations and one consolidated reference list.
- Diagrams/comparison tables where they materially improve the research argument.
- Research-to-Decision Map showing how findings should feed CivicConnect decisions.
- Contribution evidence showing meaningful participation by all three team members.
- Updated AI Usage Register where AI was used.
- Project artefacts themselves remain in the controlled CivicConnect PED/repository; do not paste a second project report into Assignment 2.

11. Responsible AI and Academic Integrity

- AI may assist brainstorming, search-term generation, explanation, comparison or drafting, but it is not an authoritative technical source.
- Verify technical claims against credible sources before relying on them.
- Do not submit fabricated citations, fabricated test results or AI-generated project evidence.
- Every team member remains accountable for understanding and defending the submitted research and recommendations.
- Record AI use according to the Master Project Brief and institutional requirements.

12. What Strong Assignment 2 Work Looks Like

- The problem is defined before a favourite solution is selected.
- Research compares alternatives rather than collecting definitions.
- The team identifies both benefits and consequences.

- Recommendations are specific to CivicConnect but are supported by research rather than preference.
- A2 evidence is used early enough to improve project decisions.
- The team does not blindly copy an A2 recommendation into M2; it still exercises engineering judgement.
- M2/PED/ADR/RTM/application evidence records the final decision and application — Assignment 2 records the research that informed it.
- Where later evidence changes a decision, the project uses controlled change rather than rewriting history.

13. Final Reminder

A2 researches — the project decides and applies

Use Assignment 2 to investigate the engineering choices your team is about to make. Use the evidence to make stronger CivicConnect decisions. Then document the final decision, rationale, traceability and application in the existing controlled project artefacts. If you are unclear about expected project standards, GitHub governance, documentation, AI accountability or evidence quality, return to the SEN381 CivicConnect Master Project Brief.